Predictive Stochastic Modeling for Financial Markets

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Abstract

This study rigorously examines the predictive power of select financial and macroeconomic series in forecasting bear market regimes within the S&P 500. By employing a two-state Markov switching autoregressive model in conjunction with both Ordinary Least Squares (OLS) and Beta Regression methodologies, we conduct extensive in-sample and out-of-sample evaluations. Our empirical findings reveal that variables such as dividend yields and total return-scaled earnings, along with macroeconomic indicators like inflation and long-term interest rates, robustly forecast bear market probabilities. These results not only challenge the semi-strong Efficient Market Hypothesis but also hold significant implications for portfolio management and monetary policy formulation.

1 Introduction

The quest to forecast financial market regimes has remained at the forefront of academic inquiry for decades. In particular, the ability to anticipate bear markets in the S&P 500 is of paramount importance—not only as a tool for optimizing portfolio allocation but also as a critical input for informed policy decisions. While the Efficient Market Hypothesis (EMH) asserts that asset prices fully reflect all publicly available information, an abundance of empirical evidence has demonstrated persistent market inefficiencies. Such inefficiencies provide the impetus for developing sophisticated models capable of capturing regime shifts and nonlinear dynamics inherent in financial time series.

In this study, we integrate a regime-switching framework with advanced predictive regression techniques to estimate the probability of bear market occurrences. Our approach builds on and extends the seminal work in the literature by incorporating both in-sample and out-of-sample tests to validate the robustness of our predictive models. The results presented herein are intended to contribute to a deeper understanding of market dynamics

and offer a rigorous methodological foundation for future research in predictive stochastic modeling.

2 Literature Review and Theoretical Background

Previous investigations (e.g., Chen, 2008; Chen, Chen, and Chou, 2017) have provided compelling evidence that the S&P 500 oscillates between distinct bull and bear regimes—with bull markets typically extending for approximately 20 months and bear markets persisting for roughly 6–7 months. Moreover, several studies have identified that financial indicators (such as dividend yields and total return-scaled earnings) and macroeconomic variables (notably inflation rates) are statistically significant predictors of market downturns.

Drawing on these insights, our research adopts a two-state Markov switching autoregressive (MS-AR) model, which adeptly captures regime-dependent features such as volatility, mean returns, and autocorrelation structures. The subsequent sections detail the data framework, econometric modeling, and empirical testing procedures that underpin our analysis.

3 Economic Framework and Data

The empirical analysis utilizes monthly data spanning January 1960 through December 2023, drawn from the U.S. Stock Markets and CAPE Ratio dataset curated by the Yale Department of Economics (Robert Shiller). All series have been adjusted for inflation. Our study focuses on the following seven key predictors:

- **Inflation (CPI)**
- **Long-Term Interest Rate** (10-year US Treasury yield)
- **Average Bond Price**
- **Dividend Yield**
- **Total Return**
- **Total Return Scaled Earnings**
- **Total Return Scaled Cyclically Adjusted Price-Earnings Ratio (TRCAPE)**

Given that unit root tests indicate nonstationarity in the raw series, we apply first differencing (with all p-values < 0.01) to ensure stationarity. The S&P 500 price series is subsequently used to compute returns and calibrate the regime-switching model.

4 Markov Switching Model

To effectively capture the dynamics of market regimes, we implement a two-state Markov switching autoregressive (MS-AR) model specified as follows:

$$r_t = \alpha_{S_t} + \mu_{S_t} + \varepsilon_{S_t},$$

where:

- $r_t = 100 \cdot \ln(\text{S\&P 500 Price})$,
- α_{S_t} and μ_{S_t} represent the regime-dependent intercept and mean, respectively,
- ε_{S_t} denotes the regime-dependent error term.

Regime transitions are governed by a fixed probability matrix:

$$P(s_t \mid s_{t-1}) = \begin{bmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{bmatrix},$$

where p_{00} and p_{11} denote the probabilities of remaining in the bear and bull regimes, respectively. Filtered probabilities, obtained via Bayesian techniques such as the Kalman filter, yield the dynamic estimate

$$Q_{0,t} = P(S_t = 0 \mid y_t),$$

which represents the probability of a bear market at time t.

For clarity, Figure 1 presents the regime-switching framework.

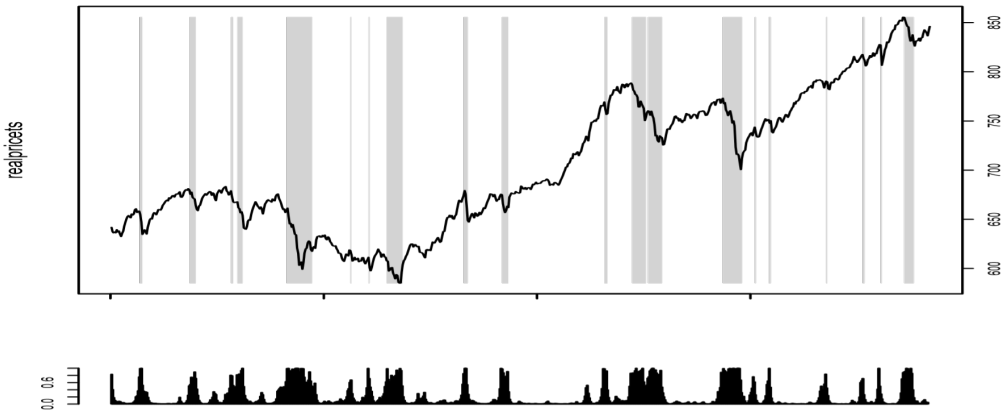


Figure 1: Regime-Switching Framework: Transition Probabilities for the S&P 500

5 Predictive Regression and In-Sample Tests

Following the estimation of regime probabilities, we evaluate the predictive capacity of each series using the regression model:

$$Q_{0,t+k} = \alpha + \beta x_t + e_{t+k},$$

where:

- $Q_{0,t+k}$ is the forecasted probability of a bear market k months ahead,
- x_t represents the predictive variable,
- e_{t+k} is the error term.

We employ both OLS and Beta Regression techniques to test the null hypothesis $\beta = 0$ (in line with the EMH) against the alternative $\beta \neq 0$.

Our in-sample tests reveal that dividend yields are statistically significant across multiple horizons (1, 2, 3, 6, 12, 18, and 24 months), while total return-scaled earnings consistently exhibit robust predictive power (with p-values below 0.01).

OLS In-Sample Results

	K=1	K=2	K=3	K=6	K=12	K=18	K=24
CPI	-0.005	-0.034**	-0.025	-0.011	0.001	-0.013	0.016
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)
Observations	767	766	765	762	756	750	744
R^2	0.0001	0.005	0.003	0.001	0.00001	0.001	0.001
Adjusted R^2	-0.001	0.004	0.001	-0.001	-0.001	-0.001	-0.0002
LRIR10	-0.111***	-0.152***	-0.103***	0.004	0.011	-0.017	-0.016
	(0.036)	(0.036)	(0.037)	(0.037)	(0.037)	(0.037)	(0.036)
Observations	767	766	765	762	756	750	744
R^2	0.012	0.022	0.010	0.00001	0.0001	0.0003	0.0003
Adjusted R^2	0.011	0.021	0.009	-0.001	-0.001	-0.001	-0.001
Real Bond	0.063***	0.054***	0.030*	-0.0003	-0.007	0.008	-0.008
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)
Observations	767	766	765	762	756	750	744
R^2	0.017	0.013	0.004	0.00000	0.0002	0.0003	0.0003
Adjusted R^2	0.016	0.011	0.003	-0.001	-0.001	-0.001	-0.001
Dividend	-0.195***	-0.206***	-0.258***	-0.325***	-0.335***	-0.189***	-0.078*
	(0.042)	(0.042)	(0.042)	(0.042)	(0.042)	(0.043)	(0.043)
Observations	767	766	765	762	756	750	744
R^2	0.027	0.030	0.047	0.075	0.079	0.026	0.005
Adjusted R^2	0.026	0.029	0.046	0.073	0.078	0.025	0.003
Total Return	-0.00000***	-0.00000	0.00000	0.00000	0.00000	0.00000	-0.00000
	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)
Observations	767	766	765	762	756	750	744
R^2	0.108	0.002	0.0001	0.00000	0.001	0.002	0.003
Adjusted R^2	0.107	0.001	-0.001	-0.001	-0.0003	0.001	0.001
TR Scaled Earnings	-0.0001***	-0.0001***	-0.00005***	-0.00003***	0.00003***	0.00003***	0.00002***
	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)
Observations	767	766	765	762	756	750	744
R^2	0.085	0.076	0.056	0.016	0.015	0.025	0.010
Adjusted R^2	0.084	0.074	0.055	0.015	0.013	0.024	0.009

Beta Regression In-Sample Results

To ensure the figure remains intact, we force the Beta Regression results onto a single page.

	K=1	K=2	K=3	K=6	K=12	K=18	K=24
CPI	-0.217***	-0.444***	-0.268***	0.003	0.035	-0.002	0.158**
	(0.072)	(0.070)	(0.071)	(0.072)	(0.072)	(0.072)	(0.072)
Observations	767	766	765	762	756	750	744
R^2	0.006	0.039	0.013	0.00000	0.0003	0.00000	0.006
LRIR10	-0.475***	-0.666***	-0.748***	0.017	-0.094	-0.040	0.021
	(0.157)	(0.156)	(0.156)	(0.158)	(0.158)	(0.158)	(0.157)
Observations	767	766	765	762	756	750	744
R^2	0.008	0.017	0.020	0.00001	0.0004	0.0001	0.00002
Real Bond	0.413***	0.426***	0.354***	-0.006	0.037	0.013	-0.103
	(0.073)	(0.073)	(0.074)	(0.074)	(0.075)	(0.074)	(0.074)
Observations	767	766	765	762	756	750	744
R^2	0.027	0.028	0.027	0.00001	0.0003	0.00004	0.002
Dividend	-0.512***	-0.480***	-0.874***	-1.642***	-1.742***	-0.686***	-0.219
	(0.184)	(0.185)	(0.184)	(0.182)	(0.181)	(0.185)	(0.185)
Observations	767	766	765	762	756	750	744
R^2	0.008	0.006	0.023	0.074	0.086	0.014	0.002
Total Return	-0.00002***	-0.00000	0.00000**	0.00000	0.00000	0.00000*	-0.00000**
	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)
Observations	767	766	765	762	756	750	744
R^2	0.249	0.001	0.004	0.0002	0.003	0.003	0.006
TR Scaled Earnings	-0.0004***	-0.0003***	-0.0003***	-0.0001***	0.0002***	0.0002***	0.0001*
	(0.00003)	(0.00003)	(0.00003)	(0.00003)	(0.00003)	(0.00003)	(0.00003)
Observations	767	766	765	762	756	750	744
R^2	0.107	0.087	0.075	0.015	0.028	0.030	0.004
TRCAPE	-0.854***	-0.072	0.101**	0.029	0.108**	0.095*	-0.058
	(0.046)	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)
Observations	767	766	765	762	756	750	744
R^2	0.291	0.002	0.004	0.0004	0.005	0.004	0.002

These in-sample results confirm that key predictors—including inflation, long-term interest rates, bond prices, dividend yields, total returns, and total return-scaled earnings—have statistically significant forecasting power.

6 Out-of-Sample Tests: Forecasting and Back-Casting

To further assess model performance, we conduct out-of-sample tests using both forecasting and back-casting exercises. For forecasting, the model is trained on data from 1960 to 2010 and then used to predict the period 2010–2023. For back-casting, the model is estimated using data from 1972 to 2023 and applied to the period 1960–1972.

We compare:

- **Restricted Model (Intercept Only):**

$$Q_{0,t+k} = \alpha_1 + e_{1,t+k},$$

- **Unrestricted Model (with Predictor):**

$$Q_{0,t+k} = \alpha_2 + \beta x_t + e_{2,t+k}.$$

Performance is measured via the Mean Squared Prediction Error (MSPE):

$$\text{MSPE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 ,$$

with an MSPE ratio (unrestricted/restricted) below 1 indicating superior forecasting accuracy.

Forecasting Results

Horizon	K=1	K=2	K=3	K=6	K=12	K=18	K=24
CPI	**0.917**	**0.658**	**0.738**	**0.860**	**0.901**	1.002	1.091
Dividend	**0.004**	**0.006**	**0.124**	**0.295**	**0.116**	**0.217**	1.013
Long Range Interest Rate	**0.982**	**0.976**	**0.976**	1.006	**0.998**	**0.993**	**0.988**
Bond Price	**0.828**	**0.822**	**0.868**	1.069	**0.952**	**0.994**	**0.963**
Total Return	**0.014**	**0.587**	**0.904**	**0.929**	**0.964**	1.303	**0.912**
TR Scaled Earnings	**0.390**	**0.412**	**0.500**	**0.764**	1.316	1.662	1.454
TR CAPE	**0.826**	**0.968**	1.013	1.012	1.028	1.032	**0.989**

Figure 4: OLS Forecasting (MSPE Ratios)

Horizon	K=1	K=2	K=3	K=6	K=12	K=18	K=24
CPI	**0.825**	**0.732**	**0.807**	**0.975**	**0.982**	1.050	1.062
Dividend	**0.495**	**0.556**	**0.389**	**0.206**	**0.244**	**0.591**	1.046
Long Range Interest Rate	**0.994**	**0.981**	**0.958**	1.001	**0.992**	**0.998**	**0.997**
Bond Price	**0.959**	**0.927**	**0.866**	1.004	**0.938**	1.002	**0.998**
Total Return	**0.442**	**0.817**	1.027	**0.888**	1.100	1.237	**0.998**
TR Scaled Earnings	**0.742**	**0.747**	**0.762**	**0.877**	1.190	1.384	1.234
TR CAPE	**0.552**	**0.986**	1.009	**0.997**	1.020	1.020	**0.998**

Figure 5: Beta Regression Forecasting (MSPE Ratios)

Back-Casting Results

Horizon	K=1	K=2	K=3	K=6	K=12	K=18	K=24
CPI	1.265	2.026	1.794	1.440	1.101	1.565	**0.356**
Dividend	1.807	1.894	2.199	2.859	2.788	2.183	1.989
Long Range Interest Rate	**0.895**	**0.854**	**0.895**	1.029	1.017	**0.990**	**0.948**
Bond Price	**0.877**	**0.891**	**0.938**	1.011	1.019	**0.979**	1.065
Total Return	1.697	1.091	**0.983**	1.002	**0.936**	**0.872**	1.321
TR Scaled Earnings	1.685	1.662	1.579	1.325	**0.737**	**0.585**	**0.512**
TR CAPE	1.239	1.023	**0.985**	**0.989**	**0.970**	**0.919**	1.141

Figure 6: OLS Back-Casting (MSPE Ratios)

Horizon	K=1	K=2	K=3	K=6	K=12	K=18	K=24
CPI	1.261	1.407	1.330	1.022	**0.982**	1.019	**0.823**
Dividend	1.093	1.085	1.160	1.255	1.227	1.133	1.055
Long Range Interest Rate	**0.966**	**0.954**	**0.938**	1.004	**0.993**	1.002	1.003
Bond Price	**0.897**	**0.915**	**0.905**	1.002	**0.995**	1.000	1.015
Total Return	**0.753**	1.014	**0.968**	**0.994**	**0.977**	**0.970**	1.042
TR Scaled Earnings	**0.942**	**0.985**	**0.997**	1.058	**0.857**	**0.889**	**0.963**
TR CAPE	**0.517**	1.005	**0.991**	**0.999**	**0.992**	**0.984**	1.017

Figure 7: Beta Regression Back-Casting (MSPE Ratios)

These out-of-sample tests demonstrate that, particularly at shorter horizons (e.g., $k = 1$ and 2), the unrestricted models incorporating predictors significantly outperform the restricted (intercept-only) models.

7 Concluding Remarks

In conclusion, this work integrates a regime-switching model with sophisticated predictive regressions to estimate bear market probabilities for the S&P 500. Our empirical analysis—spanning both in-sample and out-of-sample tests—provides compelling evidence that key financial and macroeconomic indicators, notably dividend yields and total return-scaled earnings, offer robust predictive power. These findings challenge the semi-strong form of the EMH and suggest promising avenues for enhanced risk management and market-timing strategies.

Future research should investigate the potential of advanced machine learning methods to further refine these models and capture the complex, nonlinear dynamics that characterize financial markets.

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